

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | int main(){
3 |     int num;
4 |     scanf("%d",&num);
5 |     int count=0;
6 |     count++;
7 |     for(int i=1;i<=num;++i){
8 |         count++;
9 |         count++;
10 |         if(num%i==0){
11 |             count++;
12 |         }
13 |     }
14 |     printf("%d",count);
15 |     return 0;
16 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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